FORMAL MANUSCRIPT SUBMISSION COVER LETTER

To:

Prof. Mohamed A. Abdou

Editor-in-Chief, *Fusion Engineering and Design*

Director, Fusion Science & Technology Center at UCLA

University of California, Los Angeles (UCLA), CA 90095, USA

Dear Prof. Abdou and Members of the Editorial Board,

I am pleased to submit our original research manuscript entitled **"Predicting Fast-Neutron Permeability Decay and Thermal Dryout Thresholds in Liquid Lithium Capillary Porous Systems"** for consideration as a Regular Research Article in *Fusion Engineering and Design*.

1. Alignment with Journal Scope & Core Engineering Novelty

Fusion Engineering and Design is the world-leading journal for nuclear technology, thermofluids, liquid metal MHD, and divertor engineering. This manuscript directly addresses a primary fusion engineering bottleneck: the thermal management limit of liquid metal divertors subjected to intense fast-neutron radiation.

Key novel contributions include:

- **Coupled Real Algebraic Geometry & Fluid Mechanics:** We establish a rigorous computational physics framework bounded by local energy-mass density positivity ($T_{00} >= 0$).
- **Hydrodynamic Nuance Resolution:** We resolve the counter-intuitive physics wherein neutron compaction increases Young-Laplace capillary pressure retention (ΔP_{cap} rises from 27.05 kPa to 44.48 kPa) while a 70% collapse in Carman-Kozeny permeability (K) chokes fluid resupply, driving dryout.
- **Predictive Divertor Maintenance Window:** We establish 9.8 dpa as the operational component replacement threshold (SF = 1.50) and 13.0 dpa as the complete thermal dryout parity limit (SF = 1.00).

2. Submission Declarations & Ethics

- **Exclusive Originality:** This manuscript represents original, self-contained work conducted independently by the author that has not been published previously and is not under consideration elsewhere.
- **CRediT & Ethics:** All author contributions comply with Elsevier ethical guidelines.
- **Declarations of Interest:** The author declares no competing financial or personal conflicts of interest.
- **Generative AI Usage:** Generative AI tools were used exclusively for python script plot formatting optimization. All core scientific prose, derivations, and physics were developed directly by the author.
- **Data Availability:** The complete 500-point simulation dataset, python code (cps_degradation_sim.py), high-resolution plots, and documentation are prepared for open-science repository access.

Thank you for considering this manuscript for publication in *Fusion Engineering and Design*.

Sincerely,

Jacob James Rasmussen

Independent Researcher, Shell Lake, WI 54871, USA

Incoming STEM Transfer Scholar, Chippewa Valley Technical College (CVTC), Eau Claire / Shell Lake, WI, USA